

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	03 November 2025
Team ID	NM2025TMID03627
Project Name	To Supply Leftover Food to Poor
Maximum Marks	4 Marks

Technical Architecture:

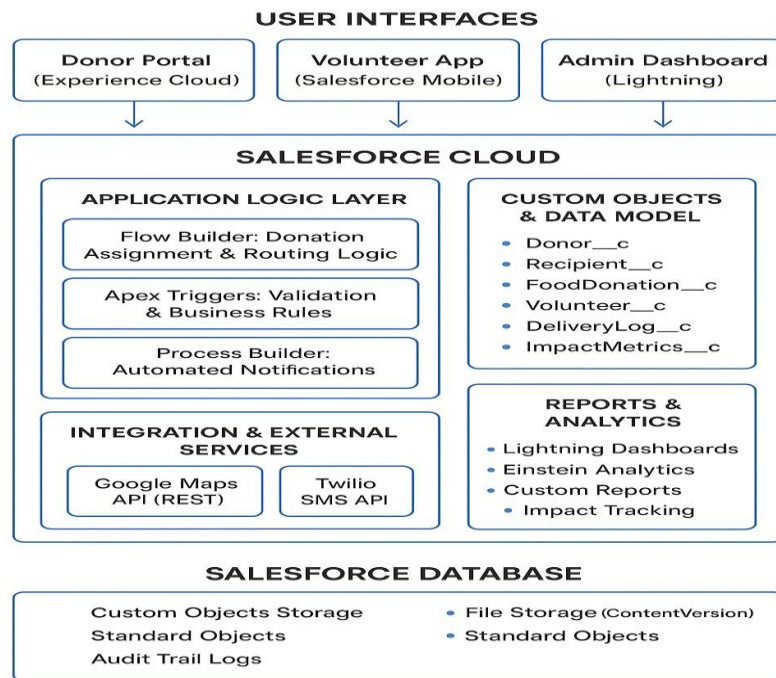


Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface - Donor	Donors submit food donation requests with details (food type, quantity, expiry, pickup location)	Salesforce Experience Cloud (Community Portal)
2.	User Interface - Volunteer	Volunteers receive notifications, accept pickups, update delivery status	Salesforce Mobile App
3.	User Interface - Admin	Admins manage donors, recipients, volunteers, and monitor system operations	Salesforce Lightning Experience
4.	Application Logic-1	Automatically assigns donations to nearest volunteers and recipients based on location and capacity	Salesforce Flow Builder, Apex
5.	Application Logic-2	Validates food donation details (expiry time, quantity) and donor credentials	Apex Triggers, Validation Rules
6.	Application Logic-3	Sends real-time notifications to volunteers about new donations	Process Builder, Salesforce Notifications
7.	Application Logic-4	Tracks donation status from creation to delivery completion	Apex Classes, Custom Controllers
8.	Application Logic-5	Calculates impact metrics (meals served, food saved, waste reduced)	Scheduled Apex Jobs, Roll-up Summary
9.	Database - Custom Objects	Stores donor, recipient, volunteer, donation, and delivery data	Salesforce Custom Objects (Donor__c, FoodDonation__c, Recipient__c, Volunteer__c, DeliveryLog__c)

S.No	Component	Description	Technology
10.	Database - Standard Objects	Uses contacts, accounts, and cases for additional tracking	Salesforce Standard Objects
11.	Cloud Database	Fully managed cloud database with automatic backups	Salesforce Cloud Database
12.	File Storage	Stores photo verification of pickups and deliveries	Salesforce Files (ContentVersion, ContentDocument)
13.	External API-1	Location-based routing and distance calculation	Google Maps API (REST)
14.	External API-2	SMS notifications to volunteers and donors	Twilio SMS API (REST)
15.	External API-3	Email notifications for status updates	SendGrid Email API / Salesforce Email Service
16.	Machine Learning Model	(Optional) Predicts donation patterns and optimizes volunteer assignment	Einstein Prediction Builder
17.	Infrastructure (Server / Cloud)	Hosted and managed on Salesforce SaaS platform with multi-tenant architecture	Salesforce Cloud (SaaS)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Not applicable (Salesforce is proprietary platform)	-
2.	Security Implementations	Role-based access control, field-level security, data encryption, OAuth 2.0 authentication	Salesforce Profiles, Permission Sets, Sharing Rules, Shield Platform Encryption

S.No	Characteristics	Description	Technology
3.	Scalable Architecture	Multi-tenant SaaS architecture, horizontally scalable with auto-scaling capabilities	Salesforce Cloud Infrastructure
4.	Availability	99.9% uptime SLA, automatic failover, disaster recovery	Salesforce Multi-Instance Architecture, Data Replication
5.	Performance	Optimized queries with SOQL, bulk processing, asynchronous operations	SOQL Query Optimization, Batch Apex, Queueable Apex, Future Methods
6.	Mobile Responsiveness	Native mobile app with offline capabilities	Salesforce Mobile SDK, Lightning Components
7.	Real-time Updates	Push notifications and live data updates	Platform Events, Streaming API
8.	Integration Capabilities	RESTful API support for third-party integrations	Salesforce REST API, SOAP API
9.	Analytics & Reporting	Real-time dashboards and custom reports for impact tracking	Lightning Reports, Dashboards, Einstein Analytics
10.	Compliance & Audit	Complete audit trail of all transactions and data changes	Field History Tracking, Setup Audit Trail